

Various statistical tests - examples

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Chi-square test

It is the Non parametric test of significance which is applied to know the association between two categorical variables.

It can be applied to know if the difference in the prevalence or proportion of an outcome in different groups is statistically significant.

Scenario	A research was conducted to know if the prevalence or proportion of hypertension among males and females in a sample population was significantly different.
Dependent variable	Hypertension – prevalence or proportion of hypertension – Categorical variable
Independent variable	Gender – categorical variable
Test of statistical significance	Chi-square test
Justification	Chi-square test is the appropriate statistical test -To study the association between two categorical variables (Here Hypertension and gender are categorical variables) -To study the difference between proportions in 2 or more different groups (Here to study the proportion / prevalence of hypertension in two different groups (two genders))

Scenario	A study was carried out to know if the prevalence of diabetes mellitus among obese and non-obese individuals were significantly different. Which test of significance is appropriate in the above situation?
Dependent variable	Prevalence or proportion or occurrence of diabetes mellitus – Categorical variable
Independent variable	Obesity – categorical variable – obese or non-obese
Test of statistical significance	Chi-square test
Justification	Chi-square test is the appropriate statistical test -To study the association between two categorical variables (Here prevalence of diabetes mellitus and obesity are categorical variables) -To study the difference between proportions in 2 or more different groups (Here to study the proportion / prevalence of diabetes mellitus in two different groups (obese and non-obese))

Independent samples T test

It is the Parametric test of significance which is applied to know if the difference between mean values in 2 independent groups are significantly different

Scenario	A research was done to know if the mean haemoglobin value among males was significantly different from mean haemoglobin value among females. The test of normality was done and it showed that data is normally distributed in the sample population among both genders.
Dependent variable	Haemoglobin values (continuous variable – interval scale)
Independent variable	Gender – categorical variable
Test of statistical significance	Independent samples T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Independent samples T test – because means in 2 independent groups or unrelated groups are compared. Here males and females are 2 independent or unrelated groups

Scenario	A research was done to know if the mean knowledge scores among employed females was significantly different from mean knowledge scores among unemployed females. The test of normality was done and it showed that data was normally distributed in the sample population among both the groups under study.
Dependent variable	Knowledge scores (continuous variable)
Independent variable	Occupation level (categorical) – Employed or unemployed
Test of statistical significance	Independent samples T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Independent samples T test – because means in 2 independent groups or unrelated groups are compared. Here employed females and unemployed females are 2 independent or unrelated groups

Paired samples T test

It is the Parametric test of significance which is applied to know if the difference between mean values in 2 related groups (before and after intervention values in the same sample) are significantly different

Scenario	A research was done to know if the mean haemoglobin value among 100 pregnant women before starting treatment with iron tablets for 3 months period was significantly different from mean haemoglobin value after treatment. The test of normality (Shapiro Wilk test) was done and it showed that data was normally distributed in the sample.
Dependent variable	Haemoglobin value in sample population (continuous variable)
Independent variable	treatment with iron tablets for 3 months period
Test of statistical significance	Paired sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Paired sample T test – because mean haemoglobin value in the sample before treatment with iron tablets for 3 months period is compared with the mean haemoglobin value in the sample after treatment. Both the values (are from the same set of patients). So it is a paired sample

Scenario	A research was done to know if the mean knowledge scores among 100 diabetic patients before a health education intervention was significantly different from mean knowledge scores among 100 diabetic patients after the health education intervention. The test of normality (Shapiro Wilk test) was done and it showed that data (knowledge scores before and after the health education intervention) was normally distributed in the sample.
Dependent variable	knowledge scores among patients (continuous variable)
Independent variable	Attending a health education intervention
Test of statistical significance	Paired sample T test
Justification	- T test because data is normally distributed (T test is a parametric test. T test is applied for comparison of means in 2 groups) - Paired sample T test – because mean knowledge scores in the sample before health education intervention is compared with mean knowledge scores in the sample after health education intervention. Both the values (are from the same set of patients). So it is a paired

ANOVA (Analysis of Variance) test

It is the Parametric test of significance which is applied to know if the difference between mean values in more than 2 independent groups are significantly different

Scenario	A research was carried out to know if the mean fasting blood sugar values among individuals from different ethnic groups (Malay, Chinese, Indians and others) were significantly different among each other. Shapiro Wilk test showed that the data was normally distributed in the sample.
Dependent variable	fasting blood sugar values among individuals (continuous variable)
Independent variable	Ethnicity (categorical variable with more than 2 groups)
Test of statistical significance	ANOVA (Analysis of Variance) Test
Justification	- ANOVA test because data is normally distributed (ANOVA is a parametric test and is applied for comparison of means in more than 2 independent groups) - Here mean FBS of individuals from different ethnic groups are compared (There are more than 2 groups).

Scenario	A research was carried out to know if the mean haemoglobin values among individuals from different socio economic class (low, middle and high) were significantly different among each other. Shapiro Wilk test showed that the data was normally distributed in the sample.
Dependent variable	Haemoglobin values among individuals (continuous variable)
Independent variable	Socio economic class (categorical variable with more than 2 groups)
Test of statistical significance	ANOVA (Analysis of Variance) Test
Justification	- ANOVA test because data is normally distributed (ANOVA is a parametric test and is applied for comparison of means in more than 2 independent groups) - Here mean haemoglobin of individuals from different socio economic class are compared (There are more than 2 groups).

Non parametric test

It is the test of significance which is applied to know if the difference between median values in 2 or more groups are significantly different.

Examples are

- **Wilcoxon Signed rank test**
- **Mann Whitney U test**
- **Kruskal Wallis test**

Scenario	A research was done to know if the mean fasting blood sugar value among 100 diabetic patients before starting treatment with anti diabetic medication for 6 months period was significantly different from mean fasting blood sugar value after treatment. The test of normality (Shapiro Wilk test) was done and it showed that data was not normally distributed in the sample.
Dependent variable	Fasting blood sugar value in sample population (continuous variable)
Independent variable	treatment with anti diabetic medication for 6 months period
Test of statistical significance	Non parametric test (Wilcoxon Signed Rank test / Mann Whitney U test)
Justification	- Non parametric test because data is not normally distributed - Comparison of means of related samples – because mean fasting blood sugar value in the sample before treatment with anti diabetic medication for 6 months period is compared with the mean fasting blood sugar value in the sample after treatment. Both the values are from the same set of patients. So it is a related sample

Scenario	A research was done to know if the mean knowledge scores among educated individuals was significantly different from mean knowledge scores among uneducated individuals. The test of normality was done and it showed that data was not normally distributed in the sample population among both the groups under study.
Dependent variable	Knowledge scores (continuous variable)
Independent variable	Education level (categorical) – Educated or uneducated
Test of statistical significance	Non parametric test for comparing two independent groups (Wilcoxon Signed Rank test / Mann Whitney U test)
Justification	- Non parametric test because data is not normally distributed - Comparison of unrelated samples was done – because 2 independent groups or unrelated groups are compared. Here educated and uneducated individuals are 2 independent or unrelated groups

Scenario	A research was carried out to know if the mean systolic blood pressure values among different occupation groups (Professionals, Clerical staff, House keeping staff) were significantly different among each other. Shapiro Wilk test showed that data was not normally distributed.
Dependent variable	systolic blood pressure values (continuous variable)
Independent variable	Occupation categories (categorical variable with more than 2 groups)
Test of statistical significance	Kruskal Wallis Test (Non parametric test)
Justification	- Kruskal Wallis test because data is not normally distributed (Kruskal Wallis is a non parametric test and is applied for comparison of means in more than 2 independent groups when data is not normally distributed) - Here mean systolic blood pressure values of individuals from different occupation groups are compared (There are more than 2 groups)